Mini Project Report - 04

Master of Computer Application – General
Semester – I

Sub: Front-End Frameworks & Technologies

Topic: Semantic Dep Website

By
Name: Anvitha S
Reg no.: PROV/ASAC/MCA/7/26/022

Faculty Name: VEERA RAGHAV K

Faculty Signature: _____

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru - 562 106

August 2026

INDEX

FIG NO	NAME OF TABLE	PG NO
01	Introduction	2
02	Input Code	3-20
03	Output Code	21-23
04	Conclusion	24

INTRODUCTION

The Semantic DEP website is a web-based platform developed to organize, manage, and present information in a structured and meaningful manner. The main purpose of the website is to make digital information easier to understand, access, search, and navigate. Unlike traditional websites that mainly display information as simple text or pages, a semantic-based website focuses on understanding the meaning and relationships between different pieces of information.

Semantic technology helps connect related data and allows information to be represented in a way that can be understood not only by users but also by computer systems. The website can use structured data, metadata, keywords, and relationships to organize its content effectively. This makes it easier for users to find relevant information and understand how different concepts are connected.

The Semantic DEP website also provides a user-friendly and interactive interface through which users can access the required information conveniently. Proper categorization and organization of data help reduce confusion and improve the overall browsing experience. The system can also support better information retrieval by allowing related information to be connected and displayed together.

INPUT CODE

```
<html>
<head>
   <title>Semantic Department</title>
   <style>
      *{
         box-sizing: border-box;
      }

      body{
         margin: 0;
         font-family: Arial, sans-serif;
         background-color: #f5f5f5;
         color: #333;
      }
      header{
         background: rgb(88, 164, 235);
         padding: 30px 20px;
         text-align: center;
         color: white;
      }
      h1{
         text-align: center;
         padding-top: 10px;
```

```
    font-family: arial, sans-serif;
    text-shadow: 2px 4px 8px rgb(0, 0, 0);
}
nav{
    display: flex;
    margin: 0px;
    padding: 0px;
    text-align: center;
    justify-content: center;
    gap: 30px;
    flex-wrap: wrap;
}
nav li{
    color: white;
    margin-top: 20px;
    margin-bottom: 20px;
    list-style-type: none;
}
nav li a{
    color: white;
    text-decoration: none;
    font-weight: bold;
}
nav li a:hover{
    color: yellow;
```

```
}
section{
    text-align: center;
    margin-top: 10px;
    padding: 30px;
}
h2{
    text-align: center;
    margin-top: 20px;
    font-size: 40px;
    font-family: franklin gothic medium, sans-serif;
    text-shadow: 2px 4px 8px gray;
}
h3{
    color: rgb(17, 0, 255);
}
article{
    display: flex;
    justify-content: space-around;
    flex-wrap: wrap;
    gap: 20px;
}
img{
    height: 150px;
    width: 150px;
```

```
        margin-top: 10px;
        border-radius: 20px;
        box-shadow: 4px 6px 8px gray;
    }
    .box{
        background-color: white;
        margin: 20px auto;
        padding: 25px;
        width: 90%;
        border-radius: 15px;
        box-shadow: 2px 4px 10px gray;
    }
    .cards{
        display: flex;
        justify-content: space-around;
        gap: 20px;
        margin-top: 70px;
        /* flex-wrap: wrap; */
    }
    .card{
        background-color: white;
        display: flex;
flex-direction: column;
        width: 280px;
        padding: 20px;
```

```
    border-radius: 15px;
    box-shadow: 3px 5px 10px gray;
}
/* .card:hover {
    transform: scale(1.03);
} */
table{
    width: 90%;
    margin: auto;
    border-collapse: collapse;
    background-color: white;
    box-shadow: 2px 4px 8px gray;
}
th{
    background-color: rgb(111, 155, 218);
    color: white;
}
th, td{
    border: 1px solid gray;
    padding: 12px;
    text-align: center;
}
marquee h2{
    color: rgb(186, 25, 108);
}
```

```
        #high {
            width: 100%;
            display: flex;
            margin: 0 auto;
            justify-content: space-around;
        }
    </style>
</head>
<body>
<header>
    <h1>Welcome to MCA Department</h1>
    <nav>
        <li><a href="#home">Home</a></li>
        <li><a href="#about">About Department</a></li>
        <li><a href="#faculty">Faculty</a></li>
        <li><a href="#courses">Courses</a></li>
        <li><a href="#timetable">Time Table</a></li>
        <li><a href="#achievements">Achievements</a></li>
        <li><a href="#events">Events</a></li>
        <li><a href="#contact">Contact</a></li>
    </nav>
</header>
<main>
<section id="home">
   <p>
```

The Master of Computer Applications (MCA) program with specializations in Data Science, General, Generative

AI, and Game Designing is designed to deliver advanced computing knowledge along with domain-specific

expertise aligned to emerging industry trends.

The program follows a comprehensive, application-driven approach, enabling students to build strong

conceptual understanding and practical proficiency in core areas such as programming (Java, Python), data

structures, algorithms, database management systems, cloud computing, and software engineering.

</p>

<marquee scrollamount="10" direction="left">

<h2>Hello MCA Graduates, Welcome to AU</h2>

</marquee>

<h3>Highlights</h3>

<article>

<img src="sunriseforever-game-console-6603120.jpg" alt="MCA Program">

<img src="paulos-ai-8330457.jpg" alt="MCA Program">

<img src="geralt-artificial-intelligence-4389372.jpg" alt="MCA Program">

<img src="elchinator-security-5043368.jpg" alt="MCA Program">

</article>

<article>

```
            <p>Generative AI</p>
            <p>Data Science</p>
            <p>Game Designing</p>
            <p>Cybersecurity</p>
        </div>
    </article>
  <div class="cards">

    <article class="card">
        <h3>Quality Education</h3>
            <p>Experienced faculty members provide quality education.</p>
    </article>
    <article class="card">
        <h3>Modern Labs</h3>
            <p>Students get practical experience through modern laboratories.</p>
    </article>
    <article class="card">
        <h3>Career Growth</h3>
        <p>Students are prepared for careers in the IT industry.</p>
    </article>
  </div>
</section>
<section id="about">
```

```
<h2>About Department</h2>
<div class="cards">
  <article class="card">
    <h3>Vision</h3>
    <p>
      To become a center of excellence in computer education,
      innovation and technological development.
    </p>
  </article>
  <article class="card">
    <h3>Mission</h3>
    <p>
      To provide quality education and practical knowledge while
      encouraging creativity, innovation and problem-solving skills.
    </p>
  </article>
  <article class="card">
    <h3>Objectives</h3>
    <p>
      To develop technical skills, programming knowledge,
      communication skills and professional abilities among students.
    </p>
  </article>
</div>
```

```
</section>
<section id="faculty">
    <h2>Our Faculty</h2>
    <table>
    <tr>
        <th>Faculty Name</th>
        <th>Designation</th>
        <th>Subject Handled</th>
        </tr>
    <tr>
        <td>Dr. Rajesh Kumar</td>
        <td>Professor</td>
        <td>Data Science</td>
        </tr>
        <tr>
            <td>Dr. Priya Sharma</td>
            <td>Associate Professor</td>
            <td>Web Development</td>
        </tr>
        <tr>
            <td>Mr. Amit Singh</td>
            <td>Assistant Professor</td>
            <td>Java Programming</td>
        </tr>
        <tr>
```

```
            <td>Ms. Neha Verma</td>
            <td>Assistant Professor</td>
            <td>Database Management</td>
        </tr>
</table>
</section>
<section id="courses">
    <h2>Courses</h2>
    <div class="cards">
        <article class="card">
            <h3>Semester 1</h3>
            <p>Programming Fundamentals</p>
            <p>Computer Networks</p>
            <p>Database Management</p>
            <p>Web Technologies</p>
        </article>
        <article class="card">
            <h3>Semester 2</h3>
            <p>Java Programming</p>
            <p>Data Structures</p>
            <p>Operating Systems</p>
            <p>Software Engineering</p>
        </article>
        <article class="card">
            <h3>Semester 3</h3>
```

```
            <p>Data Science</p>
            <p>Artificial Intelligence</p>
            <p>Machine Learning</p>
            <p>Cloud Computing</p>
        </article>
        <article class="card">
            <h3>Semester 4</h3>
            <p>Project Work</p>
            <p>Advanced Web Development</p>
            <p>Cyber Security</p>
            <p>Industry Training</p>
        </article>

    </div>
</section>
<section id="timetable">
    <h2>Class Timetable</h2>
    <table>
        <tr>
            <th>Day</th>
            <th>9:00 - 10:00</th>
            <th>10:00 - 11:00</th>
            <th>11:00 - 12:00</th>
            <th>1:00 - 2:00</th>
            <th>2:00 - 3:00</th>
```

```
</tr>
<tr>
    <td>Monday</td>
    <td>Java</td>
    <td>DBMS</td>
    <td>Web Development</td>
    <td>DSA</td>
    <td>Lab</td>
</tr>
<tr>
    <td>Tuesday</td>
    <td>DSA</td>
    <td>Java</td>
    <td>Computer Networks</td>
    <td>DBMS</td>
    <td>Lab</td>
</tr>
<tr>
    <td>Wednesday</td>
    <td>Web Development</td>
    <td>DSA</td>
    <td>Java</td>
    <td>Project</td>
    <td>Lab</td>
</tr>
```

```
    <tr>
        <td>Thursday</td>
        <td>DBMS</td>
        <td>Computer Networks</td>
        <td>Java</td>
        <td>Web Development</td>
        <td>Lab</td>
    </tr>
    <tr>
        <td>Friday</td>
        <td>Data Science</td>
        <td>DSA</td>
        <td>DBMS</td>
        <td>Project</td>
        <td>Seminar</td>
    </tr>
</table>
<h3>Lab Schedule</h3>
<table>
    <tr>
        <th>Day</th>
        <th>Lab</th>
        <th>Time</th>
    </tr>
    <tr>
```

```
            <td>Monday</td>
            <td>Java Programming Lab</td>
            <td>2:00 PM - 4:00 PM</td>
        </tr>
        <tr>
            <td>Wednesday</td>
            <td>Web Development Lab</td>
            <td>2:00 PM - 4:00 PM</td>
        </tr>

        <tr>
            <td>Friday</td>
            <td>Data Science Lab</td>
            <td>2:00 PM - 4:00 PM</td>
        </tr>
    </table>
</section>
<section id="achievements">
    <h2>Achievements</h2>
    <div class="cards">
        <article class="card">
            <h3>Student Achievements</h3>
            <p>Students participated in national-level hackathons.</p>
            <p>Students won prizes in coding competitions.</p>
            <p>Students completed industry projects.</p>
```

```
        </article>
        <article class="card">
            <h3>Faculty Achievements</h3>
            <p>Faculty members published research papers.</p>
            <p>Faculty members organized technical workshops.</p>
            <p>Faculty members guided student projects.</p>
        </article>
        <article class="card">
            <h3>Department Events</h3>
              <p>Technical fests and coding competitions are conducted regularly.</p>
            <p>Students participate in seminars and workshops.</p>
        </article>
    </div>
</section>
<section id="events">
    <h2>Events</h2>
    <div class="cards">
        <article class="card">
            <h3>Workshops</h3>
            <p>
                Workshops on Artificial Intelligence, Machine Learning,
                Web Development and Cloud Computing.
            </p>
        </article>
```

```
        <article class="card">
            <h3>Seminars</h3>
            <p>
                Expert seminars are organized to provide students with
                knowledge about current technologies.
            </p>
        </article>

        <article class="card">
            <h3>Hackathons</h3>
            <p>
                Students participate in hackathons to develop innovative
                technology-based solutions.
            </p>
        </article>
    </div>
</section>
<section id="contact">
    <h2>Contact Us</h2>
    <div class="box">
        <h3>Department Address</h3>
        <p>
            Semantic Department,<br>
            School of Computer Applications,<br>
            Alliance University,<br>
```

```
            Bengaluru, Karnataka, India
        </p>
        <p><b>Email:</b> department@example.com</p>
        <p><b>Phone:</b> +91 98765 43210</p>
    </div>

</section>
</main>
</body>
</html>
```

OUTPUT


Welcome to MCA Department
Home
About Department
Faculty
Courses
Time Table
Achievements
Events
Contact
The Master of Computer Applications (MCA) program with specializations in Data Science, General, Generative AI, and Game Designing is designed to deliver advanced computing knowledge along with domain-specific expertise aligned to emerging industry trends. The program follows a comprehensive, application-driven approach, enabling students to build strong conceptual understanding and practical proficiency in core areas such as programming (Java, Python), data structures, algorithms, database management systems, cloud computing, and software engineering.
Hello MCA Graduates, Welcome to AU
Highlights
Generative AI
Data Science
Game Designing
Cybersecurity
Quality Education
Experienced faculty members provide quality education.
Modern Labs
Students get practical experience through modern laboratories.
Career Growth
Students are prepared for careers in the IT industry.
About Department
Vision
To become a center of excellence in computer education, innovation and technological development.
Mission
To provide quality education and practical knowledge while encouraging creativity, innovation and problem-solving skills.
Objectives
To develop technical skills, programming knowledge, communication skills and professional abilities among students.

Our Faculty

Faculty Name	Designation	Subject Handled
Dr. Rajesh Kumar	Professor	Data Science
Dr. Priya Sharma	Associate Professor	Web Development
Mr. Amit Singh	Assistant Professor	Java Programming
Ms. Neha Verma	Assistant Professor	Database Management

Courses

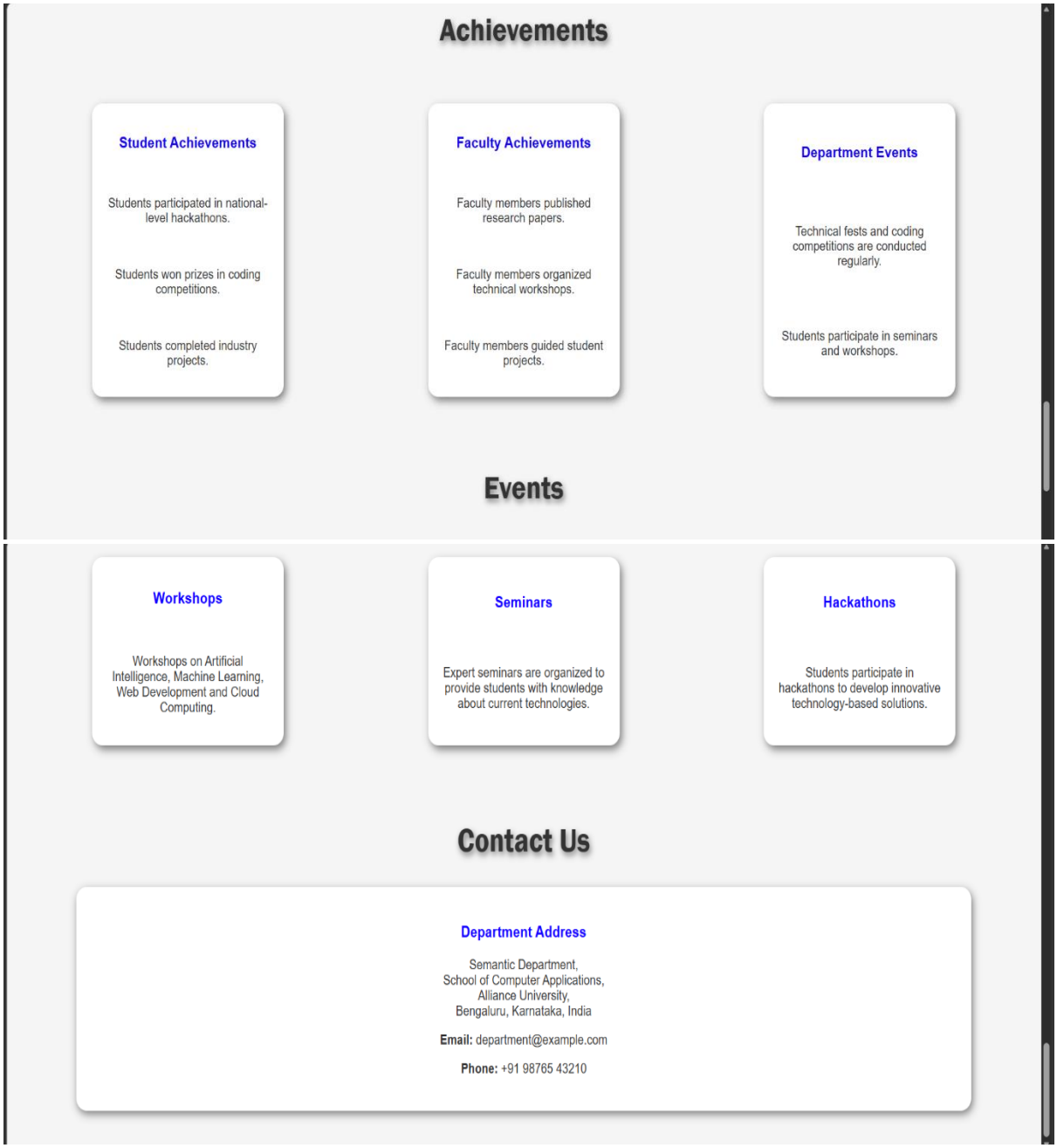
Semester 1	Semester 2	Semester 3	Semester 4
Programming Fundamentals	Java Programming	Data Science	Project Work
Computer Networks	Data Structures	Artificial Intelligence	Advanced Web Development
Programming Fundamentals	Java Programming	Data Science	Project Work
Computer Networks	Data Structures	Artificial Intelligence	Advanced Web Development
Database Management	Operating Systems	Machine Learning	Cyber Security
Web Technologies	Software Engineering	Cloud Computing	Industry Training

Class Timetable

Day	9:00 - 10:00	10:00 - 11:00	11:00 - 12:00	1:00 - 2:00	2:00 - 3:00
Monday	Java	DBMS	Web Development	DSA	Lab
Tuesday	DSA	Java	Computer Networks	DBMS	Lab
Wednesday	Web Development	DSA	Java	Project	Lab
Thursday	DBMS	Computer Networks	Java	Web Development	Lab
Friday	Data Science	DSA	DBMS	Project	Seminar

Lab Schedule

Day	Lab	Time
Monday	Java Programming Lab	2:00 PM - 4:00 PM
Wednesday	Web Development Lab	2:00 PM - 4:00 PM
Friday	Data Science Lab	2:00 PM - 4:00 PM


Achievements
Student Achievements
Students participated in national-level hackathons.
Students won prizes in coding competitions.
Students completed industry projects.
Faculty Achievements
Faculty members published research papers.
Faculty members organized technical workshops.
Faculty members guided student projects.
Department Events
Technical fests and coding competitions are conducted regularly.
Students participate in seminars and workshops.
Events
Workshops
Workshops on Artificial Intelligence, Machine Learning, Web Development and Cloud Computing.
Seminars
Expert seminars are organized to provide students with knowledge about current technologies.
Hackathons
Students participate in hackathons to develop innovative technology-based solutions.
Contact Us
Department Address
Semantic Department,
School of Computer Applications,
Alliance University,
Bengaluru, Karnataka, India
Email: department@example.com
Phone: +91 98765 43210

CONCLUSION

In conclusion, the Semantic DEP website is an effective platform that focuses on organizing, managing, and presenting information in a structured and meaningful manner. The use of semantic concepts allows different pieces of information to be connected based on their meaning and relationships, making the website more useful and easier to navigate. It helps users access relevant information quickly while providing a clear and user-friendly interface.

The website demonstrates how semantic web technologies and modern web development techniques can be combined to improve the way information is stored, connected, and retrieved. By organizing data properly, the system can reduce complexity and make information more understandable for both users and computer systems. It also provides a better method of exploring related information rather than depending only on traditional keyword-based searches.

Overall, the Semantic DEP website provides a systematic and efficient approach to information management. It improves accessibility, enhances the user experience, and shows the importance of structured and interconnected data in modern websites. With further improvements and the addition of advanced features, the website can become an even more powerful platform for managing and exploring digital information.